

Minseo Kim

Computer Vision and Robotics Research Candidate

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FICTIONAL TEST DOCUMENT - all names and details are fabricated

RESEARCH INTERESTS

3D computer vision, robot perception, visual servoing, depth sensing, SLAM, and learning from demonstration.

EDUCATION

B.S. in Computer Engineering - Hanul University (fictional)

2022-2026

- GPA: 4.12/4.50; expected graduation: February 2027.
- Relevant coursework: Computer Vision, Machine Learning, Robotics, Computer Graphics, Linear Algebra, Probability.

RESEARCH EXPERIENCE

Undergraduate Research Assistant - Vision and Intelligent Systems Lab

2025-2026

- Built an eye-to-hand camera calibration pipeline using checkerboard observations and rigid transformations.
- Processed RGB-D point clouds to estimate planar surfaces and object dimensions with RANSAC and geometric filtering.
- Evaluated pose estimation accuracy using translation and rotation error metrics across repeated trials.

Robotic Manipulation Project - Capstone Design Team

2025

- Collected 120 kinesthetic teaching trajectories from a 6-DoF robot arm and resampled each sequence to 300 steps.
- Trained a conditional autoencoder to generate action-specific joint trajectories; integrated inverse kinematics checks.
- Implemented safety validation for joint limits, maximum step size, and workspace boundaries before execution.

SELECTED PROJECTS

Depth-Based Pot Geometry Estimation - Personal Research Prototype

2026

- Estimated rim plane, bottom plane, center, radius, and height from a tilted stereo depth camera.
- Transformed measurements into a robot base frame using ArUco markers and homogeneous transformation matrices.

Visual Place Recognition - Course Project

2025

- Fine-tuned a ResNet encoder and compared cosine-similarity retrieval with handcrafted feature baselines.

TECHNICAL SKILLS

Python, C++, PyTorch, OpenCV, NumPy, SciPy, ROS 2 (basic), ZED SDK, Git, Docker (basic), Linux.

PUBLICATION / PRESENTATION

M. Kim, "Pot-Grounded Trajectory Generation for Robotic Stir-Frying," Undergraduate Research Poster, 2026. (Fictional)